Minimax Robust Designs for M-Estimated Models

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Abstract

Experimental designs that are minimax in the presence of model misspecifications have been constructed so as to minimize the maximum, over classes of alternate response models, of the integrated mean squared error of the predicted values. The theory to date has focussed almost exclusively on Least Squares estimates. Here we extend this theory to designs tailored for M-estimation of parameters, thus obtaining additional robustness against outlying, or otherwise erroneous, responses. We show that, subject to a minor change in a tuning constant, designs optimal for Least Squares remain so asymptotically for M-estimation, under squared error loss. We argue that even this minor change should be ignored, and the tuning constant chosen in an *ad hoc* but sensible manner that does not depend on which M-estimate is being employed. A rather surprising additional result is that our designs and estimates, derived under an assumption of i.i.d. errors, are also robust, in a minimax sense, against broad classes of correlation structures. We also investigate designs under a more robust loss function, less influenced by outlying responses than squared error.

Keywords: asymptotics, finite design space, Integrated M-loss, misspecified model, regression design, robustness against dependence.

1. Introduction and summary

The theory and practice of robustness of design, for possibly misspecified response functions, is well-developed as it applies to cases in which parameter estimation is to be carried out by Least Squares (LS). See Wiens (2015) for a review. An investigator seeking model robustness might naturally be concerned as well with robustness against outlying data points, or more generally against a misspecified data-generating probability distribution, and hence seek M-estimates of the parameters. Indeed, as pointed out by Müller (1997), 'a robust inference will profit from an optimal design and ... an optimal design is more reasonable if it allows also a robust analysis of the data'. See also Mizera and Müller (2001) for the special case of L_1 estimation.

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Wiens (1994, 1996) studied this design problem for quite limited classes of approximate responses on continuous design spaces and obtained asymptotic results under rather restrictive conditions. Wiens and Wu (2010) carried out a small simulation study and found that there was little apparent dependence of the designs on the method of estimation.

In this article we strengthen and extend these findings. We show that, if the parameters of the assumed model are to be estimated by ordinary M-estimation, then designs optimal, in the sense of minimizing the maximum integrated mean squared error (IMSE) of the predictions, for LS remain so, asymptotically, for M-estimation.

We note that M-estimates offer protection against outlying responses, but not necessarily against outliers in the factor space. This latter type of protection is furnished by Generalized M-estimation. But since our focus centres on design points chosen by the experimenter, such outlyingness is not an issue.

In §2.1 of this article we present the asymptotic theory on which our design problem will be based. The challenges there are somewhat unique, since we do not assume that the fitted response model is the correct one, and must allow for a broad class of alternatives. The ensuing Theorem 1 seems to be the most explicit and comprehensive response to this challenge, in the literature. In §2.2 we address the design problem. We show that the optimally robust designs, under squared error loss, depend on the anticipated method of estimation only through a tuning constant. We later argue that although this tuning constant depends on the anticipated method of estimation through unknown parameters, this dependence is so slight that it should be ignored and the tuning constant chosen in an *ad hoc* but sensible manner – as the relative emphasis placed by the designer on bias reduction versus variance reduction. The result is then that the designs are completely independent of which M-estimate is to be used.

In §2.3 we present a surprising (to us) result under which the designs and estimates presented here are also minimax robust against broad classes of correlation structures; a specific case is that in which the random errors are equicorrelated.

In §4 we consider a more robust but computationally much more expensive alternative to IMSE – 'Integrated M-loss' (IM), tailored to the robust estimation method. Whereas the 'max' part of the minimax problem addressed in §2 is carried out analytically, and in §3 we present a very efficient algorithm for the minimization, the construction of minimax IM designs requires both the 'max' and 'min' components of the problem to be carried out numerically. Algorithms for the construction of minimax IM designs are discussed in §4, with examples of both types of designs in in §5.

Proofs are in the Appendix. The MATLAB code used to prepare the examples is available on the second author's personal website.

2. Minimax robustness of design

2.1. Asymptotic theory

Our minimax design problem is phrased in terms of an approximate regression response

$$E\left[Y\left(\boldsymbol{x}\right)\right] \approx \boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}\right)\boldsymbol{\theta}, \tag{1}$$

for p regressors $\boldsymbol{f}$, each functions of vectors $\boldsymbol{x}$ of q independent variables, ranging over a finite *design space* $\chi = \{\boldsymbol{x}_1, ..., \boldsymbol{x}_N\} \subset \mathbb{R}^q$, and for a parameter vector $\boldsymbol{\theta}_{p\times 1}$.

Since (1) is an approximation, the interpretation of $\boldsymbol{\theta}$ is unclear; we *define* this target parameter by

$$\boldsymbol{\theta}_0 = \arg\min_{\boldsymbol{\eta}} \sum_{i=1}^{N} \left(E\left[Y\left(\boldsymbol{x}_i\right)\right] - \boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\boldsymbol{\eta}\right)^2. \tag{2}$$

Equivalently, and with model error $\tau\left(\boldsymbol{x}\right) \stackrel{def}{=} E\left[Y\left(\boldsymbol{x}\right)\right] - \boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}\right)\boldsymbol{\theta}_0$, we have

$$\sum_{i=1}^{N} \boldsymbol{f}\left(\boldsymbol{x}_i\right)\tau\left(\boldsymbol{x}_i\right) = \boldsymbol{0}_p. \tag{3}$$

Assuming that $\mathcal{X}$ is rich enough that the matrix $\boldsymbol{A} \stackrel{def}{=} \sum_{i=1}^{N} \boldsymbol{f}\left(\boldsymbol{x}_i\right)\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)$ is invertible, the parameter defined by (2) and (3) is unique. In order that variance and bias remain of a comparable order asymptotically, we bound the approximation error in (1) by assuming that

$$\sum_{i=1}^{N} \tau^2\left(\boldsymbol{x}_i\right) \leq \kappa^2/n, \tag{4}$$

for a constant κ. Here n is the total number of experimental runs to be carried out.

Our model is thus given by $E\left[Y\left(\boldsymbol{x}\right)\right] = \boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}\right)\boldsymbol{\theta}_0 + \tau\left(\boldsymbol{x}\right)$, for an unknown model error $\tau\left(\cdot\right)$ constrained by (3) and (4). Let Υ be the class of such functions $\tau\left(\cdot\right)$.

At such values of $\boldsymbol{x}$, observations $Y\left(\boldsymbol{x}\right)$ may be made, with additive random error: $Y\left(\boldsymbol{x}\right) = E\left[Y\left(\boldsymbol{x}\right)\right] + \varepsilon$, for i.i.d., symmetrically distributed errors ε with finite variance. We make no further distributional assumptions. In particular, the distribution of ε may be different from that assumed by the experimenter, thus calling for the type of robustness furnished by M-estimation.

Thus, given observations $\{Y_j\left(\boldsymbol{x}_i\right) | j = 1, ..., n_i\}$, with n_i replicates at distinct points $\boldsymbol{x}_i$, we suppose that $\boldsymbol{\theta}_0$ will be estimated by M-estimation with an auxiliary estimate of scale. For a function ψ with properties as in C1) below, the estimate satisfies

$$\sum_{i,j} \psi\left(\frac{Y_j\left(\boldsymbol{x}_i\right) - \boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\hat{\boldsymbol{\theta}}_n}{\hat{\sigma}_n}\right)\boldsymbol{f}\left(\boldsymbol{x}_i\right) = \boldsymbol{0}_p, \tag{5}$$

where $\hat{\sigma}_n$ is a consistent estimate of scale, for instance

$$\hat{\sigma}_n = \textsc{median}\left\{\left|Y_j\left(\boldsymbol{x}_i\right) - \boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\hat{\boldsymbol{\theta}}_n\right|\right\} \Big/ \Phi^{-1}\left(.75\right), \tag{6}$$

which is consistent for the standard deviation σ if the data are Normal. Under the mild conditions of Rocke and Shanno (1986), $\hat{\boldsymbol{\theta}}_n$ has the same asymptotic properties as if $\hat{\sigma}_n$ were replaced by σ. Thus, with $\psi_\sigma(x) \stackrel{def}{=} \psi(x/\sigma)$, we have that

$$\sum_{i,j} \psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\,\hat{\boldsymbol{\theta}}_n\right)\boldsymbol{f}(\boldsymbol{x}_i) = \mathbf{0}_p, \tag{7}$$

and we replace σ by $\hat{\sigma}_n$ in the applications. For the computations one would iterate between (5) and (6). See Huber (1981) for further details of M-estimation, Hu and Wiens (2017) for applications to testing and Hu (2018) for applications in spatial sampling.

There is a rich literature concerning minimax variance M-estimation when the true error distribution is assumed to vary over a neighbourhood of the assumed distribution – see for instance Huber (1964), Wiens (1986), Collins and Wiens (1985), Collins and Wiens (1989) for examples of solutions in various such neighbourhoods. We do not pursue this further, and leave the choice of a specific M-estimator up to the experimenter.

By a *design* we mean a probability mass function ξ on $\mathcal{X}$, to be approximated as necessary by an implementable, n-point design ξ_n, with $\xi_n(\boldsymbol{x}_i) = n_i/n$. For any design ξ define

$$\boldsymbol{M}_0(\xi) = \sum_{i=1}^{N} \boldsymbol{f}(\boldsymbol{x}_i)\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\,\xi(\boldsymbol{x}_i) \text{ and } \boldsymbol{b}_0(\xi) = \sum_{i=1}^{N} \boldsymbol{f}(\boldsymbol{x}_i)\,\tau(\boldsymbol{x}_i)\,\xi(\boldsymbol{x}_i).$$

We define $\sigma_M^2 = E\left[\psi_\sigma^2(\varepsilon)\right]/\left(E\left[\psi_\sigma'(\varepsilon)\right]\right)^2$ (assumed finite), and for a design ξ_n we set $\boldsymbol{M}_n = \boldsymbol{M}_0(\xi_n)$, and $\boldsymbol{b}_n = \boldsymbol{b}_0(\xi_n)$. In the Appendix we prove Theorem 1, stated at the end of this section, which makes precise the asymptotic normality of $\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0$:

$$\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0 \sim AN\left(\boldsymbol{M}_n^{-1}\boldsymbol{b}_n, \left(\sigma_M^2/n\right)\boldsymbol{M}_n^{-1}\right). \tag{8}$$

For this we define $\tilde{\boldsymbol{\theta}}_n = \boldsymbol{\theta}_0 + \boldsymbol{M}_n^{-1}\boldsymbol{b}_n$, and will show that $\sqrt{n}\left(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0 - \boldsymbol{M}_n^{-1}\boldsymbol{b}_n\right) = \sqrt{n}\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right)$ is asymptotically normal with mean $\mathbf{0}_p$.

For the proof of Theorem 1 we make the following assumptions and definitions. We denote the smallest and largest eigenvalues of a matrix by $ch_{\min}$ and $ch_{\max}$ respectively.

C1) The function $\psi(\cdot)$ is weakly increasing, twice differentiable and odd. We define $m_1 = \sup_t \psi_\sigma'(t)$ and $m_2 = \sup_t |\psi_\sigma''(t)|$, both assumed finite.

C2) The sequence $A_n = ch_{\min}\boldsymbol{M}_n$ is bounded away from zero.

C3) The sequence of designs has a weak limit ξ_*: $\xi_n \stackrel{d}{\to} \xi_*$ as $n \to \infty$.

Remark 1. *Assumption C1) is standard, but precludes the use of redescenders (Müller (2004)). It implies that* $E[\psi(\varepsilon)] = E[\psi''(\varepsilon)] = 0$*, since the distribution of* ε *is symmetric, and that* $E[\psi'(\varepsilon)] > 0$*, else* $\psi(\cdot)$ *is identically zero. For C2), that* $ch_{\min}\boldsymbol{M}_n >> 0$ *ensures that* $\boldsymbol{M}_n$ *remains invertible, and is a natural property of the design. For C3), that the designs have a weak limit asserts only that the design weights* $\xi_n(\boldsymbol{x}_i)$ *converge, which is a requirement of our numerical algorithm, discussed in §3. The limit will be used in the proof of Theorem 1 when we invoke Theorem 5.9 of Van der Vaart (1998), upon which we base our proof that* $\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n$ *is a* $\sqrt{n}$*-consistent estimate of zero.*

Theorem 1. *With notation as above, we have that, as* $n \to \infty$*,*

$$\sqrt{n}\boldsymbol{M}_n^{1/2}\left(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0 - \boldsymbol{M}_n^{-1}\boldsymbol{b}_n\right) \xrightarrow{d} N\left(\mathbf{0}_p, \sigma_M^2\boldsymbol{I}_p\right).$$

2.2. Minimax design theory

We define our loss in terms of the Integrated Mean Squared Error of the predictors $\hat{Y}(\boldsymbol{x}) = \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\hat{\boldsymbol{\theta}}_n$:

$$\text{IMSE}(\xi|\tau) = \sum_{\boldsymbol{x}\in\chi} E\left\{\left(E[Y(\boldsymbol{x})] - \hat{Y}(\boldsymbol{x})\right)^2\right\} = \sum_{\boldsymbol{x}\in\chi} E\left\{\left(\tau(\boldsymbol{x}) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\left(\hat{\boldsymbol{\theta}}_n(\xi|\tau) - \boldsymbol{\theta}_0(\tau)\right)\right)^2\right\}. \tag{9}$$

We will maximize an asymptotic approximation to IMSE$(\xi|\tau)$ over τ satisfying (3) and (4), and then find designs minimizing this maximum. The approximation uses the asymptotic mean and covariance given in (8). See Lehmann (1999) for a discussion limiting versus asymptotic moments.

Expanding (9) and using (3) results in

$$\text{IMSE}(\xi|\tau) = \sum_{i=1}^{N}\tau^2(\boldsymbol{x}_i) + \sum_{i=1}^{N}\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)E\left[\left(\hat{\boldsymbol{\theta}}_n(\xi|\tau) - \boldsymbol{\theta}_0(\tau)\right)\left(\hat{\boldsymbol{\theta}}_n(\xi|\tau) - \boldsymbol{\theta}_0(\tau)\right)^{\mathrm{T}}\right]\boldsymbol{f}(\boldsymbol{x}_i).$$

Substituting the asymptotic moments gives the asymptotic approximation

$$\begin{aligned}\text{AIMSE}(\xi_n|\tau) &= \sum_{i=1}^{N}\tau^2(\boldsymbol{x}_i) + \sum_{i=1}^{N}\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\left[\boldsymbol{M}_n^{-1}\boldsymbol{b}_n\boldsymbol{b}_n^{\mathrm{T}}\boldsymbol{M}_n^{-1} + \left(\sigma_M^2/n\right)\boldsymbol{M}_n^{-1}\right]\boldsymbol{f}(\boldsymbol{x}_i) \\ &= \sum_{i=1}^{N}\tau^2(\boldsymbol{x}_i) + \boldsymbol{b}_n^{\mathrm{T}}\boldsymbol{M}_n^{-1}\mathbf{A}\boldsymbol{M}_n^{-1}\boldsymbol{b}_n + \left(\sigma_M^2/n\right)tr\boldsymbol{A}\boldsymbol{M}_n^{-1}. \end{aligned} \tag{10}$$

For the maximization it is convenient to introduce an orthogonal basis for the space of regressors. Define $\boldsymbol{F}_{N\times p} = (\boldsymbol{f}(\boldsymbol{x}_1), \cdots, \boldsymbol{f}(\boldsymbol{x}_N))^{\mathrm{T}}$, and for a design ξ on χ, $\boldsymbol{D}(\xi) = diag(\xi_1, \cdots, \xi_N)$. By the Gram-Schmidt process we can construct a matrix $\boldsymbol{Q}_{N\times p}$ whose orthonormal columns form a basis for the column space of $\boldsymbol{F}$ – assumed to be of dimension p.

In the Appendix we prove the following theorem.

Theorem 2. *For any design for which $\boldsymbol{M}_0(\xi)$ in non-singular, define $p\times p$ matrices*

$$\boldsymbol{R}(\xi)=\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}(\xi)\boldsymbol{Q},\ \boldsymbol{S}(\xi)=\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}^2(\xi)\boldsymbol{Q},\ \boldsymbol{U}(\xi)=\boldsymbol{R}^{-1}(\xi)\boldsymbol{S}(\xi)\boldsymbol{R}^{-1}(\xi).$$

Then $\max_{\tau\in\Upsilon}$ AIMSE$(\xi_n|\tau)$ is given by n^{-1} times

$$J(\xi_n)=\sigma_M^2 tr\boldsymbol{R}^{-1}(\xi_n)+\kappa^2 ch_{\max}\boldsymbol{U}(\xi_n). \tag{11}$$

A further maximization of the AIMSE is discussed in the next section, after which, in §3, we minimize this maximum, thus obtaining the minimax designs.

2.3. Robustness against dependence

It is brought out in the proof of Theorem 1 – see (A.15) of the Appendix – that the estimate is representable as

$$\boldsymbol{\Phi}_n\overset{def}{=}\sqrt{n}\boldsymbol{M}_n^{1/2}\left(\hat{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0-\boldsymbol{M}_n^{-1}\boldsymbol{b}_n\right)=\boldsymbol{K}_{p\times n}\boldsymbol{u}+\boldsymbol{k}_{p\times 1}+o_p(1),$$

where $\boldsymbol{K}$ and $\boldsymbol{k}$ are non-random and $\boldsymbol{u}=(U_1,...,U_n)^{\mathrm{T}}$ with $U_i=\psi_\sigma(\varepsilon_i)/E[\psi'_\sigma(\varepsilon)]$. We have up to now assumed that the errors ε_i are i.i.d., so that as well the U_i are i.i.d., with $\mathrm{COV}[\boldsymbol{u}]=\sigma_M^2\boldsymbol{I}_n$ and, asymptotically, $\mathrm{COV}[\boldsymbol{\Phi}_n]=\sigma_M^2\boldsymbol{K}\boldsymbol{K}^{\mathrm{T}}$. We now investigate the effect on the estimate if the U_i are correlated, or are heteroscedastic.

Suppose then that $\mathrm{COV}[\boldsymbol{u}]=\boldsymbol{C}_{n\times n}$, so that the asymptotic covariance structure of $\boldsymbol{\Phi}_n$ becomes $\mathrm{COV}[\boldsymbol{\Phi}_n|\boldsymbol{C}]=\boldsymbol{K}\boldsymbol{C}\boldsymbol{K}^{\mathrm{T}}$. Note that $\mathrm{COV}[\boldsymbol{\Phi}_n|\boldsymbol{C}]$ is non-decreasing in the Loewner ordering: $\boldsymbol{C}_1\preceq\boldsymbol{C}_2\Rightarrow\mathrm{COV}[\boldsymbol{\Phi}_n|\boldsymbol{C}_1]\preceq\mathrm{COV}[\boldsymbol{\Phi}_n|\boldsymbol{C}_2]$. If loss is measured by $\mathcal{L}(\boldsymbol{C})=\phi(\mathrm{COV}[\boldsymbol{\Phi}_n|\boldsymbol{C}])$, where $\phi(\cdot)$ is a function, such as the trace, determinant, maximum eigenvalue etc. that is itself non-decreasing in the Loewner ordering, then $\mathcal{L}(\boldsymbol{C})$ is non-decreasing in this ordering. In the particular problem at hand, $\phi(\Sigma)=\sum_{i=1}^N\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\Sigma\boldsymbol{f}(\boldsymbol{x}_i)$ is the integrated variance of the predictors when evaluated at $\boldsymbol{\Sigma}=\mathrm{COV}[\Phi_n|\boldsymbol{C}]$.

Suppose now that $\|\cdot\|_M$ is a matrix norm, induced by a vector norm $\|\cdot\|_V$, i.e. $\|\boldsymbol{C}\|_M=\sup_{\|x\|_V=1}\|\boldsymbol{C}\boldsymbol{x}\|_V$. Special cases are the spectral radius $\|\boldsymbol{C}\|_E$ – this is the maximum eigenvalue since $\boldsymbol{C}$ is a covariance matrix – and the maximum absolute row sum $\|\boldsymbol{C}\|_1=\max_i\sum_j|c_{ij}|$.

The following lemma is given in Wiens (2025); we repeat it here for convenience.

Lemma 1. *For $\eta^2>0$, covariance matrix $\boldsymbol{C}$ and induced matrix norm $\|\boldsymbol{C}\|_M$, define*

$$\mathcal{C}_M=\left\{\boldsymbol{C}\mid\boldsymbol{C}\succeq 0\ \textit{and}\ \|\boldsymbol{C}\|_M\le\eta^2\right\}.$$

For the norm $\|\cdot\|_E$ an equivalent definition is $\mathcal{C}_E=\{\boldsymbol{C}\mid 0\preceq\boldsymbol{C}\preceq\eta^2\boldsymbol{I}_n\}$. Then (i) in any such class $\mathcal{C}_M$, $\max_{\mathcal{C}_M}\mathcal{L}(\boldsymbol{C})=\mathcal{L}(\eta^2\boldsymbol{I}_n)$, and (ii) if $\mathcal{C}'\subseteq\mathcal{C}_M$ and $\eta^2\boldsymbol{I}_n\in\mathcal{C}'$, then $\max_{\mathcal{C}'}\mathcal{L}(\boldsymbol{C})=\mathcal{L}(\eta^2\boldsymbol{I}_n)$.

A consequence of (i) of this lemma is that if one is carrying out a statistical procedure with loss function $\mathcal{L}(\boldsymbol{C})$, then a version of the procedure that minimizes $\mathcal{L}(\eta^2 \boldsymbol{I}_n)$ is *minimax* as $\boldsymbol{C}$ varies over $\mathcal{C}_M$. By (ii) this remains true for subsets of $\mathcal{C}_M$ that contain $\eta^2 \boldsymbol{I}_n$. To apply this result we need only ensure that η^2 is large enough that $\mathcal{C}_M$ contains the departures, from independence or homoscedasticity, that are of interest. If so, the designs and estimates of this article – i.e. those derived under an assumption of i.i.d. errors – enjoy the additional optimality property of minimizing the maximum value of $\mathcal{L}(\boldsymbol{C})$, as the covariance matrix $\boldsymbol{C}$ of $\boldsymbol{u}$ varies over $\mathcal{C}_M$.

Lemma 1 is unsatisfactory in this M-estimation context since it is based on the covariance structure of $\{\psi_\sigma(\varepsilon_i)\}$, rather than that of $\{\varepsilon_i\}$. In general there is no tractable relationship between the two. There are however important exceptions. Suppose that under the dependence structure the $\{\varepsilon_i\}$ are *exchangeable*, hence *equicorrelated*. Then the $\{\psi_\sigma(\varepsilon_i)\}$ also have this structure: for some ρ (typically *not* the correlation among the $\{\varepsilon_i\}$), and some α^2,

$$\text{COV}[\boldsymbol{u}] = \boldsymbol{C} = \alpha^2\left((1-\rho)\boldsymbol{I}_n + \rho \mathbf{1}_n \mathbf{1}_n^{\mathrm{T}}\right).$$

We impose bounds $|\rho| \leq \rho_{\max} < 1$ and $\alpha^2 \leq \alpha^2_{\max}$. We shall work with the norm $\|\boldsymbol{C}\|_1 = \alpha^2(1+(n-1)|\rho|)$, this is also $\|\boldsymbol{C}\|_E$ if $\rho \geq 0$. Set $\eta^2 = \alpha^2_{\max}(1+(n-1)\rho_{\max})$. Then $\|\boldsymbol{C}\|_1 \leq \eta^2$ and Lemma 1 applies: an assumption of i.i.d. errors with $\text{VAR}[U_i] = \eta^2$, which for convenience we write as $\eta^2 = \eta_0^2\sigma_M^2$, is minimax within a class containing all such equicorrelated error structures.

By a similar treatment one can show robustness against dependence structures under which ε_i and ε_j are independent for $|i-j| > q$, as is the case for $MA(q)$ errors. The case of independent but heteroscedastic errors is clearly covered as well, by $\alpha^2_{\max} \geq \max_i \text{VAR}[\psi_\sigma(\varepsilon_i)]$.

3. Construction of minimax AIMSE designs

With

$$\nu \stackrel{def}{=} \kappa^2 / \left(\eta_0^2\sigma_M^2 + \kappa^2\right), \tag{12}$$

the results of §2.2 easily accommodate those of §2.3: the maximum, over both $\tau \in \Upsilon$ and $\boldsymbol{C} \in \mathcal{C}_1$, of (10) is given by $\left(\eta_0^2\sigma_M^2 + \kappa^2\right)/n$ (which does not depend on the design) times

$$I_\nu(\xi_n) = (1-\nu)\, tr\boldsymbol{R}^{-1}(\xi_n) + \nu ch_{\max}\boldsymbol{U}(\xi_n). \tag{13}$$

If robustness against dependence is not of interest then set $\eta_0^2 = 1$.

A *minimax* AIMSE design is a minimizer of $I_\nu(\xi_n)$. For fixed ν, $I_\nu(\xi_n)$ is precisely the value minimized, in Wiens (2018), to obtain minimax designs for LS estimates, thus justifying our statement in §1 that such designs remain minimax optimal, asymptotically, for M-estimation.

The minimization of $I_\nu(\xi_n)$ is carried out sequentially, as described in Theorem 5 of Wiens (2018). Briefly, given a current k-point design ξ_k, the loss resulting from the addition of a design point at $\boldsymbol{x}_i$ is expanded as

$$I_\nu\left(\xi_{k+1}^{(i)}\right) = I_\nu(\xi_k) - t_{k,i}/k + O\left(k^{-2}\right), \tag{14}$$

and then $\boldsymbol{x}_{(i)}$, with $(i) = \arg\max_i t_{k,i}$, is added to the design. This is carried out to convergence, yielding a design ξ on χ with intended allocations $n_i = n\xi(\boldsymbol{x}_i)$. Typically most $\xi(\boldsymbol{x}_i)$ are zero, but otherwise the n_i are not integers. To obtain implementable designs ξ_n we first round up the n_i to $\lceil n\xi(\boldsymbol{x}_i)\rceil$, whose sum exceeds n. The excess is decreased stepwise, by removing points whose value of $t_{k,i}$ in (14) is a minimum. This method typically results in only a very small increase in the minimized value of $I_\nu(\xi_n)$.

4. Integrated M-loss

Our method of measuring loss – IMSE – is itself heavily influenced by outlying responses, and hence is non-robust. As an alternative, in this section we consider *Integrated M-loss* (IM), in which IMSE is replaced by

$$\text{IM}(\xi,\tau) = \sum_{\boldsymbol{x}\in\boldsymbol{\chi}} E\left[\rho\left(E\left[Y(\boldsymbol{x})\right] - \hat{Y}(\boldsymbol{x})\right)\right] = \sum_{\boldsymbol{x}\in\boldsymbol{\chi}} E\left[\rho\left(\tau(\boldsymbol{x}) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\left(\hat{\boldsymbol{\theta}}_n(\xi|\tau) - \boldsymbol{\theta}_0(\tau)\right)\right)\right],$$

where $\rho(x) = \int_0^x \psi(t)\,dt$. This is evaluated at the asymptotic distribution (8), yielding 'Asymptotic Integrated M-loss' (AIM). If ψ is Huber's score function then IM is termed 'Huber loss' (Hastie et al. (2009), p. 349). If $\psi(x) = x$ then IM is identical to IMSE applied to LS estimates. We continue to impose constraints (3) and (4) on $\tau(\boldsymbol{x})$. We define $\boldsymbol{H}(\boldsymbol{\xi}_n) = \left(\boldsymbol{I}_N - \boldsymbol{Q}\boldsymbol{R}^{-1}(\boldsymbol{\xi}_n)\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}(\boldsymbol{\xi}_n)\right)\boldsymbol{Q}_\perp : N \times N-p$, where $\boldsymbol{R}(\boldsymbol{\xi})$ is as in Theorem 2 and, as in the proof of that theorem, $\boldsymbol{Q}_\perp : N \times N-p$ is the orthogonal complement of $\boldsymbol{Q}$.

Lemma 2. *Asymptotic Integrated M-loss is given by* $\text{AIM} = \max_{\|\boldsymbol{\beta}\|\le 1} G_0(\boldsymbol{\beta})$*, for*

$$G_0(\boldsymbol{\beta}) = \sum_{i=1}^{N}\int_{-\infty}^{\infty}\rho\left(\frac{\kappa}{\sqrt{n}}\boldsymbol{h}_i^{\mathrm{T}}(\boldsymbol{\xi}_n)\boldsymbol{\beta} - \frac{\sigma_M}{\sqrt{n}}s_i(\boldsymbol{\xi}_n)z\right)\phi(z)\,dz, \tag{15}$$

where $\boldsymbol{h}_i^{\mathrm{T}}(\boldsymbol{\xi}_n)$ *is the* i^{th} *row of* $\boldsymbol{H}(\boldsymbol{\xi}_n)$*,* $s_i(\boldsymbol{\xi}_n) = \sqrt{\left(\boldsymbol{Q}\boldsymbol{R}^{-1}(\boldsymbol{\xi}_n)\boldsymbol{Q}^{\mathrm{T}}\right)_{ii}}$*, and* $\phi(z)$ *is the standard Normal density.*

The computation of minimax AIM designs is expensive, since the maximization of $G_0(\boldsymbol{\beta})$, done analytically for IMSE, must now be done numerically. For this we employ MATLAB's constrained minimizer `fmincon` to minimize $-G_0(\boldsymbol{\beta})$ over $\|\boldsymbol{\beta}\| \le 1$. This is sped up by supplying the gradient and Hessian, and by vectorizing the input functions. The integrals are approximated via Gauss-Hermite quadrature. In Lemma 3 below these integrals are presented in the form $I = \int_{-\infty}^{\infty} g(\boldsymbol{\eta}(z))\,\phi(z)dz$ for a vectorized function g. This is an $N \times 1$ vector with i^{th} element $\int_{-\infty}^{\infty} g(\eta_i(z))\,\phi(z)\,dz$. This vector of integrals can be approximated very efficiently as $\hat{I} = g(\boldsymbol{\eta}(\boldsymbol{z}^{\mathrm{T}}))\,\boldsymbol{w}$, where the matrix $g(\boldsymbol{\eta}(\boldsymbol{z}^{\mathrm{T}}))$ has $(i,j)^{th}$ element $g(\eta_i(z_j))$, for Gauss-Hermite quadrature nodes z_j, with corresponding weights w_j in $\boldsymbol{w}$.

Lemma 3. *Assume that ρ, ψ and ψ' are vectorized, so that if $\boldsymbol{M}$ is a matrix then $\rho(\boldsymbol{M})$ is a matrix of the same size with $[\rho(\boldsymbol{M})]_{ij} = \rho(\boldsymbol{M}_{ij})$, and similarly with ψ and ψ'. Let $\boldsymbol{s}(\boldsymbol{\xi}_n)$ have elements $s_i(\boldsymbol{\xi}_n)$ and define*

$$\boldsymbol{\eta}(\boldsymbol{\xi}_n, z) = \frac{\kappa}{\sqrt{n}}\boldsymbol{H}(\boldsymbol{\xi}_n)\boldsymbol{\beta} - \frac{\sigma_M}{\sqrt{n}}\boldsymbol{s}(\boldsymbol{\xi}_n)z : N \times 1.$$

Then

$$G_0(\boldsymbol{\beta}) = \mathbf{1}_N^{\mathrm{T}} \int_{-\infty}^{\infty} \rho(\boldsymbol{\eta}(\boldsymbol{\xi}_n, z))\phi(z)\,dz, \tag{16}$$

with $N - p \times 1$ gradient

$$\boldsymbol{G}_1(\boldsymbol{\beta}) = \frac{\kappa}{\sqrt{n}}\boldsymbol{H}^{\mathrm{T}}(\boldsymbol{\xi}_n) \int_{-\infty}^{\infty} \psi(\boldsymbol{\eta}(\boldsymbol{\xi}_n, z))\phi(z)\,dz, \tag{17}$$

and $N - p \times N - p$ Hessian

$$\boldsymbol{G}_2(\boldsymbol{\beta}) = \frac{\kappa^2}{n}\boldsymbol{H}^{\mathrm{T}}(\boldsymbol{\xi}_n) \oplus_{i=1}^{N} \left[\int_{-\infty}^{\infty} \psi'(\boldsymbol{\eta}(\boldsymbol{\xi}_n, z))\phi(z)\,dz\right]_i \boldsymbol{H}(\xi_n). \tag{18}$$

The subsequent minimization of $\max_{\|\boldsymbol{\beta}\|\leq 1} G_0(\boldsymbol{\beta})$ over $\boldsymbol{\xi}_n$ is also carried out by `fmincon`; this is quite time-consuming since each trial design $\boldsymbol{\xi}_n$ must be accompanied by a maximizing $\boldsymbol{\beta}$. This latter point renders the sequential approach to the minimization, as employed in §3, untenable due to the huge number of maximizations required. The final design returned by `fmincon` is made implementable by applying the 'Efficient Design Apportionment' method of Pukelsheim and Rieder (1992).

5. Examples, implementations and discussion

5.1. Minimax AIMSE designs

To obtain the minimax AIMSE design only ν, at (13), need be specified by the user. The simplest and most natural way to do this is to view ν as expressing the emphasis on the reduction of losses due to bias, rather than to variation. Its choice is then up to the user; quite typically $\nu = .5$ is chosen. In this method there is *no* difference in the minimax designs for different M-estimates.

Another method is suggested by the definition of ν at (12). Although the parameters involved in this definition would not be known to the designer, it is of interest to see how their values could affect the resulting designs.

We begin by seeing how much ν can change from its value under LS. The relationship between ν using Least Squares and ν using the M-estimate is, with $\gamma \stackrel{def}{=} \eta_0\sigma/\kappa$, that

$$\nu_{\mathrm{LS}} = \left(\gamma^2 + 1\right)^{-1} \text{ and } \nu_{\mathrm{M}} = \left(\frac{\gamma^2\sigma_M^2}{\sigma^2} + 1\right)^{-1}.$$

We assess these assuming that $\varepsilon/\sigma \sim N(0,1)$ and that $\psi(x) = xI(|x| \leq c) + sign(x)cI(|x| > c)$ (Huber (1964)).

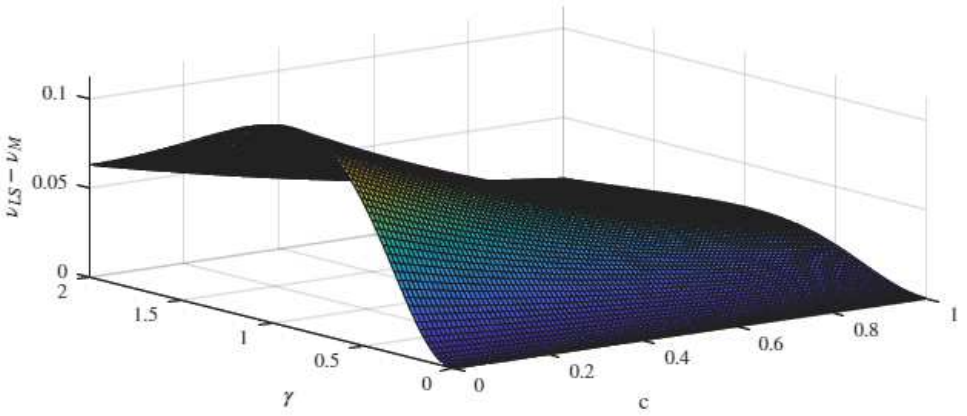


Figure 1: Differences $\nu_{\rm LS} - \nu_{\rm M}$ in terms of γ and c.

Lemma 4. *With notation as above,*

$$0 \le \nu_{LS} - \nu_M \le \frac{\sqrt{\pi/2} - 1}{\sqrt{\pi/2} + 1} \simeq .1124.$$

The lower bound is attained only when the M-estimate is the LSE*, and the upper bound is attained when the M-estimate is the L_1 estimate and $\gamma^2 = 1/\sqrt{\pi/2} \simeq .7979$. At the maximum*

$$\nu_{LS} = \frac{\sqrt{\pi/2}}{\sqrt{\pi/2} + 1} \simeq .5562, \quad \nu_M = \frac{1}{\sqrt{\pi/2} + 1} = 1 - \nu_{LS} \simeq .4438. \tag{19}$$

See Figure 1. We have constructed designs, using the 'worst case' values $\nu_{\rm LS}$ and $\nu_{\rm M}$ of (19), and also their midpoint $\nu = .5$. See Figure 2 for approximate linear regression and Figure 3 for approximate cubic regression, each on design spaces consisting of 20 equally spaced points spanning $[-1, 1]$. Given that $\nu_{\rm LS}$, $\nu_{\rm M}$ and the M-estimate were chosen so as to make the designs as different as possible, these designs are remarkably similar – even identical after being made implementable. The implementations all can be described as taking the replicates that would otherwise be assigned by the classically ($\nu = 0$) I-optimal designs and spreading them out into clusters at nearby design points. The I-optimal design for linear regression places mass of .5 at each of ± 1. That for cubic regression was derived by Studden (1977) and places masses of .1545 and .3455 at ± 1 and $\pm .4472$.

We conclude that an experimenter measuring loss by IMSE should feel quite safe in using the same design for an experiment regardless of which M-estimate is to be employed, and in choosing ν to represent his desired emphasis on bias reduction, as posited at the beginning of this section. As well, the assumption of i.i.d. errors is safe, even optimal in a minimax sense, amongst equicorrelated or heteroscedastic error structures (and others).

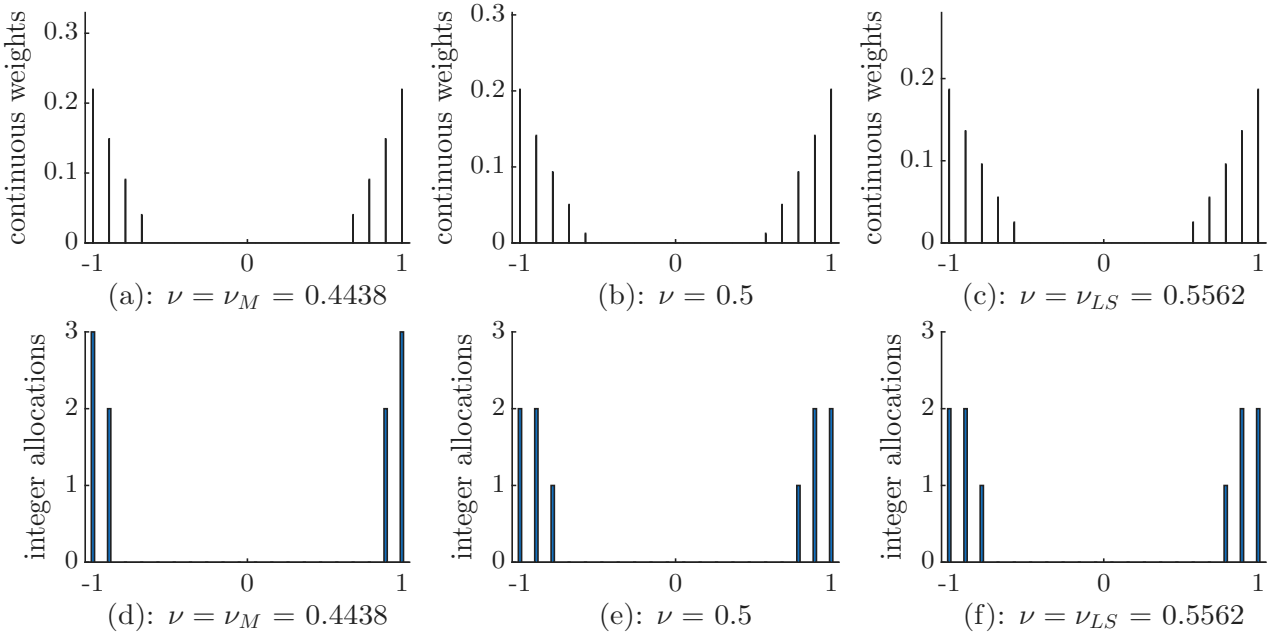


Figure 2: Designs for linear regression; n=10, N=20. (a)-(c): Continuous weights; (d)-(f): Integer allocations.

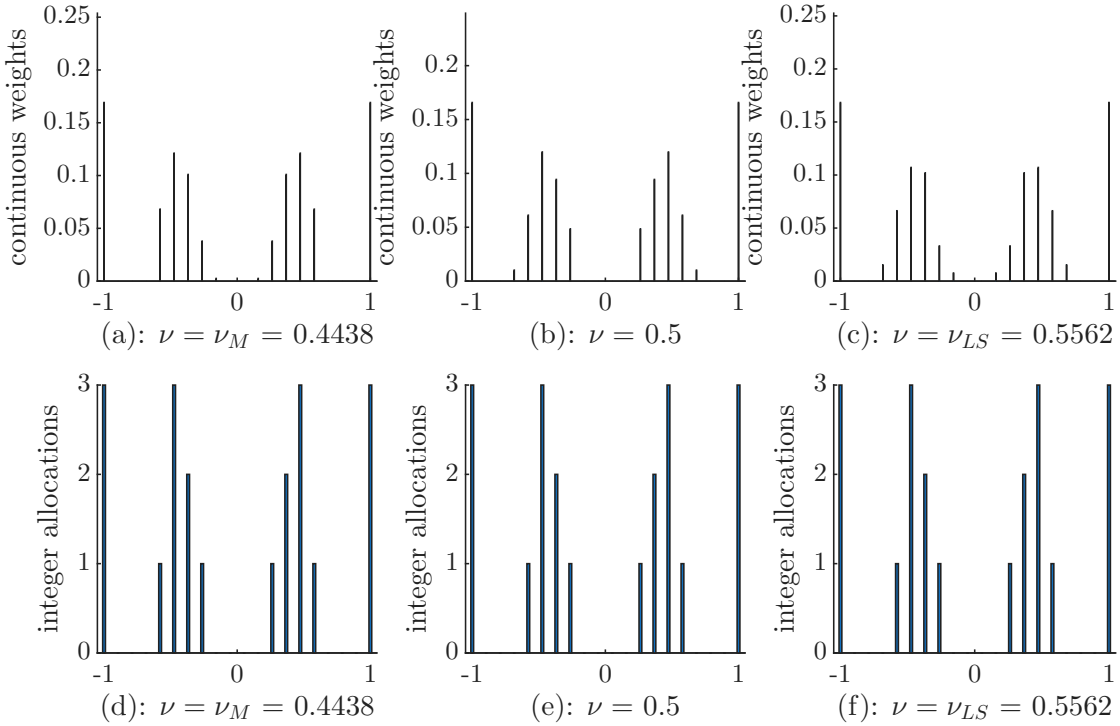


Figure 3: Designs for cubic regression; n=20, N=20. (a)-(c): Continuous weights; (d)-(f): Integer allocations.

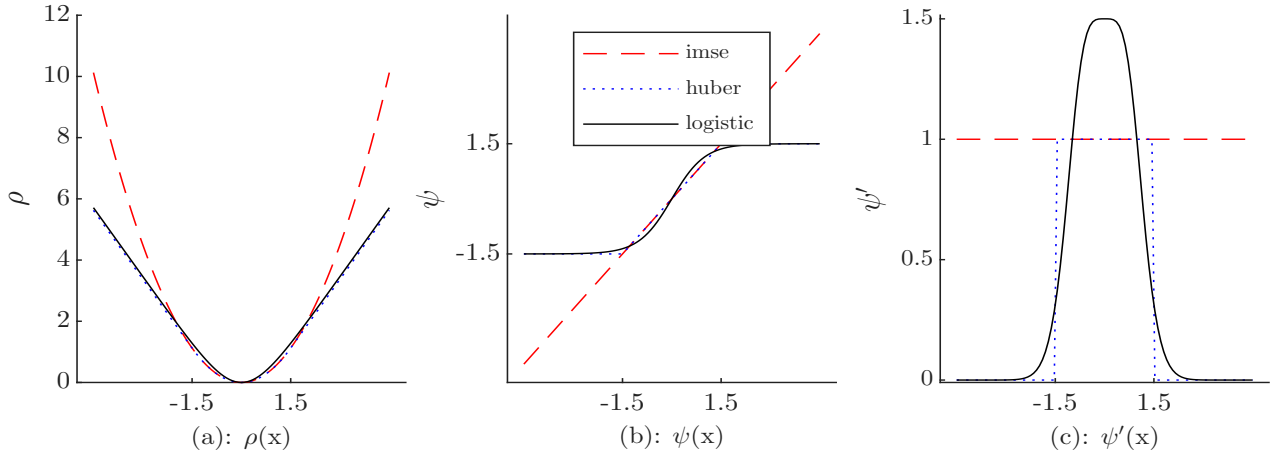


Figure 4: Loss functions $\rho(x)$ with their derivatives for IM designs. IMSE loss: $\psi(x) = x$; Huber loss: $\psi(x) = xI(|x| \leq 1.5) + sign(x)1.5I(|x| > 1.5)$; logistic loss: $\psi(x) = 1.5\tanh(x)$.

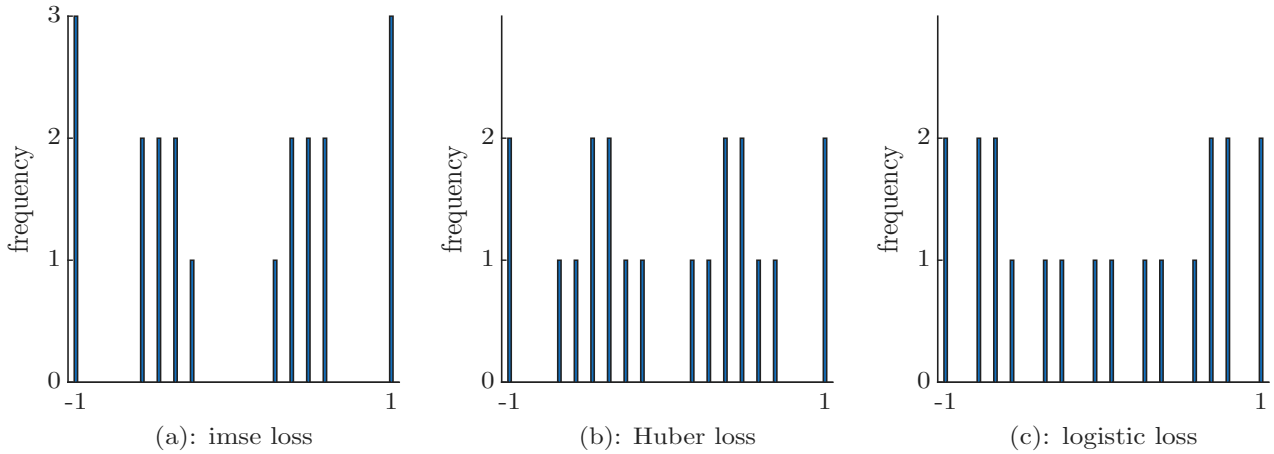


Figure 5: Minimax AIM designs for cubic regression; n=20, N=20.

5.2. Minimax AIM designs

We have applied the procedure described in §4 using the three loss functions illustrated in Figure 4. Figure 5 gives the results for approximate cubic regression, using the same values of n and N as in Figure 3, and $\kappa = \sigma_M = 5$ (hence $\nu = .5$). The function being minimized has local minima, and we had to run the algorithm repeatedly, using a different starting design to initialize the iterative process each time, before arriving at what appears to be at least very close to an absolute minimum. The designs are qualitatively very similar to those of 3(e), the differences in the clusters arising from the different implementation method being used, as well as the robust loss functions used in Figures 5(b) and 5(c), which typically result in more diffuse designs. This latter feature is quite pronounced in Figure 6, in which the fitted model is linear in two variables $x_1, x_2 \in [-1, 1]$. The designs in Figures 6(a) and 6(b) are based on squared error loss and, as expected, have clusters of design

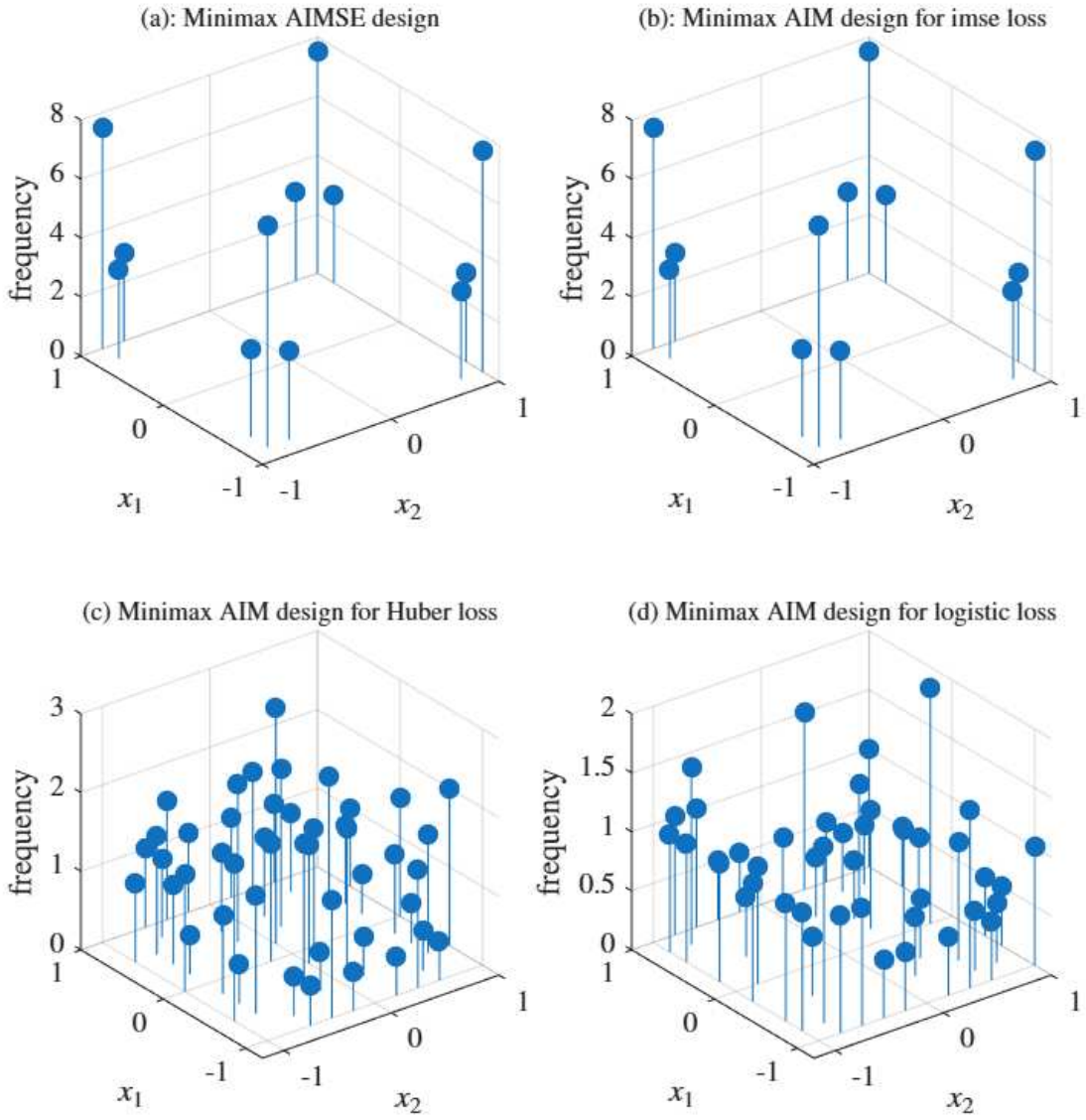


Figure 6: Minimax AIM designs for bivariate linear regression; n=36, N=121.

points in the corners of the design space. The robustness of the designs in Figures 6(c) and 6(d) is reflected in the increased spread of these clusters.

Appendix: Proofs

Remark 2. *It is convenient to identify a design with its design measure – a discrete probability measure $\xi_n(d\boldsymbol{x})$ on $\mathcal{X}$. Thus, as an example, if the design calls for n_i runs to be made at $\boldsymbol{x}_i \in \chi$, we write averages of functions ϕ of the data as*

$$\frac{1}{n}\sum_{\{i|n_i>0\}}\sum_{j=1}^{n_i}\phi\left(Y_j\left(\boldsymbol{x}_i\right)\right)=\sum_{\{i|n_i>0\}}\frac{n_i}{n}\bar{\phi}\left(\boldsymbol{x}_i\right)=\int_{\mathcal{X}}\bar{\phi}\left(\boldsymbol{x}\right)\xi_n\left(d\boldsymbol{x}\right),$$

where $\bar{\phi}\left(\boldsymbol{x}_i\right)=\left(1/n_i\right)\sum_{j=1}^{n_i}\phi\left(Y_j\left(\boldsymbol{x}_i\right)\right)$. We abbreviate $\sum_{\{i|n_i>0\}}\sum_{j=1}^{n_i}$ by $\sum_{i,j}$.

Applying the theorem of Van der Vaart (1998) mentioned in Remark 1 requires that we verify further properties of the limit, given in Lemma A.1 below.

Lemma A.1. *As well as C1) - C3) assume that the parameter space Θ is a compact subset of $\mathbb{R}^p$. Then with $\Psi_n\left(\boldsymbol{\theta}\right)\overset{def}{=}n^{-1}\sum_{i,j}\psi_\sigma\left(Y_j\left(\boldsymbol{x}_i\right)-\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\left(\boldsymbol{\theta}+\tilde{\boldsymbol{\theta}}_n\right)\right)\boldsymbol{f}\left(\boldsymbol{x}_i\right)$, the function*

$$\Psi\left(\boldsymbol{\theta}\right)=\int_{\chi}E\left[\psi_\sigma\left(Y\left(\boldsymbol{x}\right)-\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}\right)\left(\boldsymbol{\theta}+\boldsymbol{\theta}_0\right)\right)\right]\boldsymbol{f}\left(\boldsymbol{x}\right)\xi_*\left(d\boldsymbol{x}\right)$$

satisfies

$$\begin{array}{ll}(i) & \sup_{\boldsymbol{\theta}\in\boldsymbol{\Theta}}\left\|\Psi_n\left(\boldsymbol{\theta}\right)-\Psi\left(\boldsymbol{\theta}\right)\right\|\overset{pr}{\to}0,\\ (ii) & \text{for every } \delta>0,\ \inf_{\|\boldsymbol{\theta}\|\geq\delta}\left\|\Psi\left(\boldsymbol{\theta}\right)\right\|>0=\left\|\Psi(\mathbf{0}_p)\right\|.\end{array}$$

Proof: With $\tilde{\boldsymbol{\Psi}}_n(\boldsymbol{\theta})\overset{def}{=}\int_{\chi}E\left[\psi_\sigma\left(Y(\boldsymbol{x})-\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})(\boldsymbol{\theta}+\widetilde{\boldsymbol{\theta}}_n)\right)\right]\boldsymbol{f}(\boldsymbol{x})\,\xi_n(d\boldsymbol{x})=E\left[\Psi_n\left(\boldsymbol{\theta}\right)\right]$ we shall establish (i) of the Lemma by showing:

(a) $\sup_{\boldsymbol{\theta}\in\boldsymbol{\Theta}}\left\|\tilde{\boldsymbol{\Psi}}_n\left(\boldsymbol{\theta}\right)-\boldsymbol{\Psi}\left(\boldsymbol{\theta}\right)\right\|\to 0$,
(b) $\sup_{\boldsymbol{\theta}\in\boldsymbol{\Theta}}\left\|\boldsymbol{\Psi}_n\left(\boldsymbol{\theta}\right)-\tilde{\boldsymbol{\Psi}}_n\left(\boldsymbol{\theta}\right)\right\|\overset{pr}{\to}0$.

For (a), with $\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\eta})\overset{def}{=}E[\psi_\sigma(Y(\boldsymbol{x})-\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})(\boldsymbol{\theta}+\boldsymbol{\eta}))]\,\boldsymbol{f}(\boldsymbol{x})$ we have $\boldsymbol{\Psi}(\boldsymbol{\theta})=\int_{\chi}\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\theta}_0)\,\xi_*(d\boldsymbol{x})$ and $\tilde{\boldsymbol{\Psi}}_n\left(\boldsymbol{\theta}\right)=\int_{\chi}\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\tilde{\boldsymbol{\theta}}_n)\,\xi_n(d\boldsymbol{x})$, with

$$\begin{aligned}\boldsymbol{\Psi}(\boldsymbol{\theta})-\tilde{\boldsymbol{\Psi}}_n\left(\boldsymbol{\theta}\right) &= \int_{\chi}\left(\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\theta}_0)-\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\tilde{\boldsymbol{\theta}}_n)\right)\xi_n(d\boldsymbol{x})+\int_{\chi}\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\theta}_0)\,(\xi_*-\xi_n)(d\boldsymbol{x})\\ &= \boldsymbol{\alpha}_n(\boldsymbol{\theta})+\boldsymbol{\beta}_n(\boldsymbol{\theta}),\text{ say.}\end{aligned}$$

Since $\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\eta})$ is differentiable w.r.t. $\boldsymbol{\eta}$, and both ψ'_σ and $\boldsymbol{f}$ are bounded, there is $C>0$ for which $\left\|\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\theta}_0)-\boldsymbol{h}(\boldsymbol{x},\boldsymbol{\theta},\tilde{\boldsymbol{\theta}}_n)\right\|\leq C\left\|\boldsymbol{\theta}_0-\tilde{\boldsymbol{\theta}}_n\right\|$, uniformly in $\boldsymbol{x}\in\chi$ and $\boldsymbol{\theta}\in\Theta$. Hence

$$\sup_{\boldsymbol{\theta}\in\Theta}\left\|\boldsymbol{\alpha}_n(\boldsymbol{\theta})\right\|\leq C\left\|\boldsymbol{\theta}_0-\tilde{\boldsymbol{\theta}}_n\right\|\to 0. \tag{A.1}$$

With $\boldsymbol{g}_{\boldsymbol{\theta}}(\boldsymbol{x}) \stackrel{def}{=} \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{\theta}, \boldsymbol{\theta}_0)$ we obtain, in a similar fashion, $\sup_{\boldsymbol{x}\in\chi} \|\boldsymbol{g}_{\boldsymbol{\theta}_1}(\boldsymbol{x}) - \boldsymbol{g}_{\boldsymbol{\theta}_2}(\boldsymbol{x})\| \leq C\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|$. Then since $\boldsymbol{\Theta}$ is compact, for every $\delta > 0$ there exist $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(K)} \in \Theta$ such that every $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ lies within distance δ of one of these points. Therefore,

$$\sup_{\boldsymbol{\theta}\in\Theta} \|\boldsymbol{\beta}_n(\boldsymbol{\theta})\| \leq 2C\delta + \max_{1\leq k\leq K} \left\| \int_{\chi} \boldsymbol{g}_{\boldsymbol{\theta}^{(k)}}(\boldsymbol{x})\, \xi_*(d\boldsymbol{x}) - \int_{\chi} \boldsymbol{g}_{\boldsymbol{\theta}^{(k)}}(\boldsymbol{x})\, \xi_n(d\boldsymbol{x}) \right\|.$$

That $\xi_n \stackrel{d}{\to} \xi_*$ implies that the preceding norms, hence their maximum, converge to 0. Letting $\delta \to 0$, we conclude that $\sup_{\boldsymbol{\theta}\in\Theta} \|\boldsymbol{\beta}_n(\boldsymbol{\theta})\| \to 0$, which together with (A.1) yields (a).

For (b), we write $\boldsymbol{\Delta}_n(\boldsymbol{\theta}) \stackrel{def}{=} \boldsymbol{\Psi}_n(\boldsymbol{\theta}) - \tilde{\boldsymbol{\Psi}}_n(\boldsymbol{\theta}) = n^{-1}\sum_{i,j} \left\{ \boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta}) - E[\boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta})] \right\}$, where

$$\boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta}) = \psi_\sigma\Big(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\,(\boldsymbol{\theta} + \tilde{\boldsymbol{\theta}}_n)\Big)\, \boldsymbol{f}(\boldsymbol{x}_i).$$

Arguing as above, there exists $C > 0$ such that for all $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \boldsymbol{\Theta}$ and each pair (i, j),

$$\left\| \boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta}_1) - \boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta}_2) \right\| \leq C\, \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|.$$

Taking expectations gives the same bound for the centred version, so that

$$\|\boldsymbol{\Delta}_n(\boldsymbol{\theta}_1) - \boldsymbol{\Delta}_n(\boldsymbol{\theta}_2)\| \leq 2C\, \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|.$$

Thus Δ_n is uniformly Lipschitz on $\boldsymbol{\Theta}$, with a constant independent of n.

Fix $\varepsilon > 0$. Choose $\delta > 0$ such that $2C\delta < \varepsilon/2$, and let $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(K)}$ be a finite δ-net of Θ, as above. Then

$$\sup_{\boldsymbol{\theta}\in\Theta} \|\boldsymbol{\Delta}_n(\boldsymbol{\theta})\| \leq \max_{1\leq k\leq K} \left\| \boldsymbol{\Delta}_n(\boldsymbol{\theta}^{(k)}) \right\| + \varepsilon/2,$$

and it remains only to show that

$$\max_{1\leq k\leq K} \left\| \boldsymbol{\Delta}_n(\boldsymbol{\theta}^{(k)}) \right\| \stackrel{pr}{\to} 0. \tag{A.2}$$

Write $\Delta_n(\boldsymbol{\theta}^{(k)}) = n^{-1}\sum_{i,j} \left(\boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta}^{(k)}) - E\left[\boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta}^{(k)}) \right] \right) = n^{-1}\sum_{i,j} \boldsymbol{Z}_{ij,n}^{(k)}$, say. We can write $\boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta})$ as $\boldsymbol{\zeta}_{ij,n}(\boldsymbol{\theta}) = \psi_\sigma(\varepsilon_{ij} + e_{i,n}(\boldsymbol{\theta}))\, \boldsymbol{f}(\boldsymbol{x}_i)$, for certain functions $e_{i,n}(\boldsymbol{\theta})$, continuous on $\boldsymbol{\Theta}$ hence uniformly bounded there. Then $E[\boldsymbol{Z}_{ij,n}^{(k)}] = \boldsymbol{0}$ and there is $C > 0$ for which $\sup_{i,j,n,k} E\|\boldsymbol{Z}_{ij,n}^{(k)}\|^2 \leq C$. Since the $\boldsymbol{Z}_{ij,n}^{(k)}$ are independent for each (k, n), $\mathrm{VAR}\left[\Delta_n(\boldsymbol{\theta}^{(k)})\right] \leq n^{-1}C \to 0$, implying by Chebyshev's inequality that $\boldsymbol{\Delta}_n(\boldsymbol{\theta}^{(k)}) \stackrel{pr}{\to} 0$ for each $k = 1, \ldots, K$. Now (A.2) follows, thus completing the proof of (b).

For part (ii) of the Lemma, recall that the design space is finite and that each $\xi_n(\boldsymbol{x}_i)$ tends to $\xi_*(\boldsymbol{x}_i)$, so that

$$\int_{\chi} E\left[\psi_\sigma\left(Y(\boldsymbol{x}) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\,\boldsymbol{\theta}_0\right)\right] \boldsymbol{f}(\boldsymbol{x})\, \xi_*(d\boldsymbol{x}) = \lim_{n\to\infty} \int_{\chi} E\left[\psi_\sigma\left(\varepsilon + \tau(\boldsymbol{x})\right)\right] \boldsymbol{f}(\boldsymbol{x})\, \xi_n(d\boldsymbol{x}) = \boldsymbol{0}_p, \tag{A.3}$$

since $\tau(\boldsymbol{x})$ is $O\left(n^{-1/2}\right)$. It follows that $\Psi(\mathbf{0}_p) = \mathbf{0}_p$. By (A.3), $\boldsymbol{\theta}_0$ is a stationary point of the function $k(\boldsymbol{\theta}) \stackrel{def}{=} \int_\chi E\left[\rho_\sigma\left(Y(\boldsymbol{x}) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\boldsymbol{\theta}\right)\right]\xi_*(d\boldsymbol{x})$, where $\rho_\sigma(x) = \int_0^x \psi_\sigma(t)\,dt$. The Hessian is $\ddot{k}(\boldsymbol{\theta}) = \int_\chi E\left[\psi'_\sigma\left(\varepsilon + M_{\boldsymbol{\theta}}(\boldsymbol{x})\right)\right]\boldsymbol{f}(\boldsymbol{x})\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\,\xi_*(d\boldsymbol{x})$, for $M_{\boldsymbol{\theta}}(\boldsymbol{x}) = E[Y(\boldsymbol{x})] - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\boldsymbol{\theta}$). Since ψ_σ is weakly increasing, $\psi'_\sigma\left(\varepsilon + M_{\boldsymbol{\theta}}(\boldsymbol{x})\right) \geq 0$ and so $E\left[\psi'_\sigma\left(\varepsilon + M_{\boldsymbol{\theta}}(\boldsymbol{x})\right)\right]$ must be strictly positive, else $\psi'_\sigma\left(\varepsilon + M_{\boldsymbol{\theta}}(\boldsymbol{x})\right) \equiv 0$ (a.s.). Put

$$c(\boldsymbol{\theta}) = \min_{i=1,\ldots,N}\left\{E\left[\psi'_\sigma\left(\varepsilon + M_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\right)\right] \mid \xi_*(\boldsymbol{x}_i) > 0\right\}.$$

Then $c(\boldsymbol{\theta}) > 0$ and $\ddot{k}(\boldsymbol{\theta}) \succeq c(\boldsymbol{\theta})\int_\chi \boldsymbol{f}(\boldsymbol{x})\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\,\xi_*(d\boldsymbol{x}) \succ \mathbf{0}$, so that $\boldsymbol{\theta}_0$ is the unique stationary point of the strictly convex function $k(\boldsymbol{\theta})$ and $\mathbf{0}_p$ is the unique zero of $\|\boldsymbol{\Psi}(\boldsymbol{\theta})\|$.

Since $\boldsymbol{\Theta}$ is compact, its closed subset $\boldsymbol{\Theta}_\delta$ defined by $\|\boldsymbol{\theta}\| \geq \delta$ is also compact, hence the continuous function $\|\boldsymbol{\Psi}(\boldsymbol{\theta})\|$ attains its inf there:

$$\inf_{\boldsymbol{\Theta}_\delta}\|\boldsymbol{\Psi}(\boldsymbol{\theta})\| = \min_{\boldsymbol{\Theta}_\delta}\|\boldsymbol{\Psi}(\boldsymbol{\theta})\| > 0.$$

This proves (ii), and completes the proof of Lemma A.1. □

A.1. Proof of Theorem 1

We must show that $\sqrt{n}\boldsymbol{M}_n^{1/2}\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right) \xrightarrow{d} N(\mathbf{0}_p, \sigma_M^2\boldsymbol{I}_p)$. The stochastic component of the estimate will be expressed in terms of

$$\bar{\boldsymbol{z}}_n \stackrel{def}{=} n^{-1}\sum_{i,j}\psi_\sigma(\varepsilon_{ij})\,\boldsymbol{f}(\boldsymbol{x}_i). \tag{A.4}$$

Note that $\bar{\boldsymbol{z}}_n$ is an average of independent r.v.s, each with mean zero, and covariance matrices

$$\textsc{var}\left[\psi_\sigma(\varepsilon)\right]\boldsymbol{f}(\boldsymbol{x}_i)\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) = E\left[\psi_\sigma^2(\varepsilon)\right]\boldsymbol{f}(\boldsymbol{x}_i)\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i),$$

so that $\textsc{cov}[\bar{\boldsymbol{z}}_n] = n^{-1}E\left[\psi_\sigma^2(\varepsilon)\right]\boldsymbol{M}_n \succ \mathbf{0}_p$. By the Lindberg-Feller CLT,

$$\boldsymbol{M}_n^{-1/2}\sqrt{n}\bar{\boldsymbol{z}}_n \xrightarrow{d} N\left(\mathbf{0}_p, E\left[\psi_\sigma^2(\varepsilon)\right]\boldsymbol{I}_p\right). \tag{A.5}$$

Completing the proof of Theorem 1 requires a preliminary Lemma.

Lemma A.2. *Recall that* $\tilde{\boldsymbol{\theta}}_n = \boldsymbol{\theta}_0 + \boldsymbol{M}_n^{-1}\boldsymbol{b}_n$. *We have that* $\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n$ *is a* $\sqrt{n}$*-consistent estimate of* $\mathbf{0}_p$*:*

$$\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n = O_p\left(n^{-1/2}\right). \tag{A.6}$$

Proof: We use the functions

$$\begin{aligned} a_n(\boldsymbol{x}) &= \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\left(\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\right) = \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\boldsymbol{M}_n^{-1}\boldsymbol{b}_n, \\ h_n(\boldsymbol{x}) &= \tau(\boldsymbol{x}) - a_n(\boldsymbol{x}), \\ d_n(\boldsymbol{x}) &= \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right). \end{aligned}$$

By (4) and C2), both $a_n(\boldsymbol{x})$ and $h_n(\boldsymbol{x})$ are $o_p(1)$.

We first show that $\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n$ is a consistent estimate of $\mathbf{0}_p$:

$$\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n = o_p(1). \tag{A.7}$$

We have that $\Psi_n\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right) = n^{-1}\sum_{i,j}\psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\hat{\boldsymbol{\theta}}_n\right)\boldsymbol{f}(\boldsymbol{x}_i) = \mathbf{0}_p$, by (7). Now Theorem 5.9 of Van der Vaart (1998), with $\boldsymbol{\theta}_0 = \mathbf{0}_p$ and under the conditions there – ensured by our Lemma A.1 – asserts (A.7), and also that $d_n(\boldsymbol{x}) = o_p(1)$.

We require the stronger $\sqrt{n}$-consistency. For this, observe that by (7),

$$\begin{aligned}
\mathbf{0}_p &= \frac{1}{n}\sum_{i,j}\psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\hat{\boldsymbol{\theta}}_n\right)\boldsymbol{f}(\boldsymbol{x}_i) \\
&= \frac{1}{n}\sum_{i,j}\psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\tilde{\boldsymbol{\theta}}_n - d_n(\boldsymbol{x}_i)\right)\boldsymbol{f}(\boldsymbol{x}_i) \\
&= \frac{1}{n}\sum_{i,j}\psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\tilde{\boldsymbol{\theta}}_n\right)\boldsymbol{f}(\boldsymbol{x}_i) - \boldsymbol{G}_n\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right),
\end{aligned} \tag{A.8}$$

where for generic $t_i \in [0,1]$ and $u_{ij} \in [0,1]$, and with $s_{n,i} = h_n(\boldsymbol{x}_i) - t_i d_n(\boldsymbol{x}_i) = o_p(1)$,

$$\begin{aligned}
\boldsymbol{G}_n &= \frac{1}{n}\sum_{i,j}\psi'_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\tilde{\boldsymbol{\theta}}_n - t_i d_n(\boldsymbol{x}_i)\right)\boldsymbol{f}(\boldsymbol{x}_i)\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) \\
&= \frac{1}{n}\sum_{i,j}\psi'_\sigma(\varepsilon_{ij} + s_{n,i})\boldsymbol{f}(\boldsymbol{x}_i)\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) \\
&= \frac{1}{n}\sum_{i,j}\psi'_\sigma(\varepsilon_{ij})\boldsymbol{f}(\boldsymbol{x}_i)\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) + \frac{1}{n}\sum_{i,j}\psi''_\sigma(\varepsilon_{ij} + u_{ij}s_{n,i})s_{n,i}\boldsymbol{f}(\boldsymbol{x}_i)\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i).
\end{aligned}$$

We will show that $\boldsymbol{G}_n$ tends in probability to the positive definite matrix $\boldsymbol{G}_* = E\left[\psi'_\sigma(\varepsilon)\right]\int_\chi \boldsymbol{f}(\boldsymbol{x})\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\,\xi_*(d\boldsymbol{x})$. Then by (A.8), and ignoring terms which are $o_p(n^{-1/2})$,

$$\begin{aligned}
\sqrt{n}\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right) &= \left(\boldsymbol{G}_*^{-1} + o_p(1)\right)\left\{\frac{1}{\sqrt{n}}\sum_{i,j}\psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\tilde{\boldsymbol{\theta}}_n\right)\boldsymbol{f}(\boldsymbol{x}_i)\right\} \\
&= \left(\boldsymbol{G}_*^{-1} + o_p(1)\right)\left\{\frac{1}{\sqrt{n}}\sum_{i,j}\psi_\sigma\left(\varepsilon_{ij} + h_n(\boldsymbol{x}_i)\right)\boldsymbol{f}(\boldsymbol{x}_i)\right\} \\
&= \left(\boldsymbol{G}_*^{-1} + o_p(1)\right)\left\{\sqrt{n}\bar{\boldsymbol{z}}_n + \frac{1}{n}\sum_{i,j}\psi'_\sigma\left(\varepsilon_{ij} + t_i h_n(\boldsymbol{x}_i)\right)\sqrt{n}h(\boldsymbol{x}_i)\boldsymbol{f}(\boldsymbol{x}_i)\right\},
\end{aligned}$$

where the first term in the final set of braces is $O_p(1)$ by (A.5), and the second is $O_p(1)$ because

$$\sqrt{n}h_n(\boldsymbol{x}) = \sqrt{n}\tau(\boldsymbol{x}) - \sqrt{n}\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\boldsymbol{M}_n^{-1}\boldsymbol{b}_n = \sqrt{n}\tau(\boldsymbol{x}) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\boldsymbol{M}_n^{-1}\int_\chi \boldsymbol{f}(\boldsymbol{x})\,\sqrt{n}\tau(\boldsymbol{x})\,\xi_n(d\boldsymbol{x})$$

is bounded, again by virtue of (4). Thus showing that

$$\boldsymbol{G}_n \xrightarrow{pr} \boldsymbol{G}_*, \tag{A.9}$$

will complete the verification of (A.6).

To show (A.9) note that by the triangle inequality,

$$\|\boldsymbol{G}_n - \boldsymbol{G}_*\| \leq \left\|\boldsymbol{G}_n - \frac{1}{n}\sum_{i,j}\psi'_\sigma(\varepsilon_{ij})\,\boldsymbol{f}(\boldsymbol{x}_i)\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\right\| + \|\boldsymbol{W}_n\| + \left\|\frac{1}{n}\sum_{i,j}E\left[\psi'_\sigma(\varepsilon_{ij})\right]\boldsymbol{f}(\boldsymbol{x}_i)\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) - \boldsymbol{G}_*\right\|, \tag{A.10}$$

where

$$\boldsymbol{W}_n \overset{def}{=} \frac{1}{n}\sum_{i,j}\left(\psi'_\sigma(\varepsilon_{ij}) - E\left[\psi'_\sigma(\varepsilon_{ij})\right]\right)\boldsymbol{f}(\boldsymbol{x}_i)\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) = \sum_i \xi_n(\boldsymbol{x}_i)\,V_i\boldsymbol{f}(\boldsymbol{x}_i)\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i),$$

for independent r.v.s $V_i = \sum_{j=1}^{n_i}\left(\psi'_\sigma(\varepsilon_{ij}) - E\left[\psi'_\sigma(\varepsilon_{ij})\right]\right)/n_i$ with means 0 and variances $\mathrm{VAR}[\psi'_\sigma(\varepsilon)]/n_i \leq m_1^2/n_i$.

The first norm on the rhs of (A.10) is

$$\left\|\frac{1}{n}\sum_{i,j}\psi''_\sigma(\varepsilon_{ij} + u_{ij}s_{n,i})\,s_{n,i}\boldsymbol{f}(\boldsymbol{x}_i)\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\right\| \leq \frac{m_2}{n}\sum_{i,j}|s_{n,i}|\,\|\boldsymbol{f}(\boldsymbol{x}_i)\|^2 = o_p(1),$$

and the third is $\left\|\int_\chi E\left[\psi'_\sigma(\varepsilon)\right]\boldsymbol{f}(\boldsymbol{x})\,\boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x})\left(\xi_n - \xi_*\right)(d\boldsymbol{x})\right\| = o(1)$. A typical element of W_n is $W_{n,k,l} = \sum_i \xi_n(\boldsymbol{x}_i)\,V_i f_k(\boldsymbol{x}_i)\,f_l(\boldsymbol{x}_i)$, with mean zero and

$$\mathrm{VAR}\left[W_{n,k,l}\right] = \sum_i \frac{n_i^2}{n^2}\mathrm{VAR}\left[V_i\right]\left(f_k(\boldsymbol{x}_i)\,f_l(\boldsymbol{x}_i)\right)^2 \leq \frac{m_1^2}{n}\int_\chi \left(f_k(\boldsymbol{x}_i)\,f_l(\boldsymbol{x}_i)\right)^2\xi_n(d\boldsymbol{x}) = O(n^{-1}),$$

so that by Chebyshev's Inequality, $W_{n,k,l} \xrightarrow{pr} 0$ and so $\boldsymbol{W}_n \xrightarrow{pr} \boldsymbol{0}$. Now (A.9) follows, completing the proof of the Lemma. □

We can now complete the proof of Theorem 1. To establish the asymptotic normality of $\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n$, we observe that by (7),

$$\begin{aligned}
\boldsymbol{0}_p &= \frac{1}{n}\sum_{i,j}\psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\,\hat{\boldsymbol{\theta}}_n\right)\boldsymbol{f}(\boldsymbol{x}_i) \\
&= \frac{1}{n}\sum_{i,j}\psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\,\tilde{\boldsymbol{\theta}}_n - d_n(\boldsymbol{x}_i)\right)\boldsymbol{f}(\boldsymbol{x}_i) \\
&= \frac{1}{n}\sum_{i,j}\psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\,\tilde{\boldsymbol{\theta}}_n\right)\boldsymbol{f}(\boldsymbol{x}_i) \\
&\quad -\frac{1}{n}\sum_{i,j}\psi'_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\,\tilde{\boldsymbol{\theta}}_n\right)\boldsymbol{f}(\boldsymbol{x}_i)\,d_n(\boldsymbol{x}_i) + \boldsymbol{S}_n,
\end{aligned} \tag{A.11}$$

where

$$\boldsymbol{S}_n = (1/2n)\sum_{i,j}\psi''_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i)\,\tilde{\boldsymbol{\theta}}_n + t_i d_n(\boldsymbol{x}_i)\right)d_n^2(\boldsymbol{x}_i)\,\boldsymbol{f}(\boldsymbol{x}_i).$$

With $B_n \stackrel{def}{=} ch_{\max} \int_\chi \boldsymbol{f}(\boldsymbol{x}) \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}) \|\boldsymbol{f}(\boldsymbol{x})\| \xi_n(d\boldsymbol{x}) \leq \int_\chi \|\boldsymbol{f}(\boldsymbol{x})\|^3 \xi_n(d\boldsymbol{x}) < \infty$, we have

$$\|\boldsymbol{S}_n\| \leq \frac{m_2}{2n} \sum_{i,j} \left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right)^{\mathrm{T}} \boldsymbol{f}(\boldsymbol{x}_i) \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) \left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right) \|\boldsymbol{f}(\boldsymbol{x}_i)\| \leq \frac{m_2}{2} B_n \left\|\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right\|^2 = O_p\left(n^{-1}\right).$$

The final step in the preceding line uses Lemma A.2. Thus, from (A.11) and disregarding terms that are $O_p\left(n^{-1/2}\right)$,

$$\sqrt{n}\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right) = \boldsymbol{P}_n^{-1} \frac{1}{\sqrt{n}} \sum_{i,j} \psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) \tilde{\boldsymbol{\theta}}_n\right) \boldsymbol{f}(\boldsymbol{x}_i), \tag{A.12}$$

where

$$\begin{aligned} \boldsymbol{P}_n &\stackrel{def}{=} \frac{1}{n} \sum_{i,j} \psi'_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) \tilde{\boldsymbol{\theta}}_n\right) \boldsymbol{f}(\boldsymbol{x}_i) \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) \\ &= \frac{1}{n} \sum_{i,j} \psi'_\sigma\left(\tau(\boldsymbol{x}_i) + \varepsilon_{ij} - a_n(\boldsymbol{x}_i)\right) \boldsymbol{f}(\boldsymbol{x}_i) \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) \\ &= \frac{1}{n} \sum_{i,j} \psi'_\sigma\left(\tau(\boldsymbol{x}_i) + \varepsilon_{ij}\right) \boldsymbol{f}(\boldsymbol{x}_i) \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) - \boldsymbol{T}_n, \end{aligned} \tag{A.13}$$

and

$$\boldsymbol{T}_n = \frac{1}{n} \sum_{i,j} \psi''_\sigma\left(\tau(\boldsymbol{x}_i) + \varepsilon_{ij} - t_i a_n(\boldsymbol{x}_i)\right) a_n(\boldsymbol{x}_i) \boldsymbol{f}(\boldsymbol{x}_i) \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i).$$

For any vector $\boldsymbol{\alpha}$ of unit norm,

$$\begin{aligned} |\boldsymbol{\alpha}^{\mathrm{T}} \boldsymbol{T}_n \boldsymbol{\alpha}| &\leq \frac{1}{n} \sum_{i,j} \left|\psi''_\sigma\left(\tau(\boldsymbol{x}_i) + \varepsilon_{ij} - t a_n(\boldsymbol{x}_i)\right) a_n(\boldsymbol{x}_i)\right| \left(\boldsymbol{\alpha}^{\mathrm{T}} \boldsymbol{f}(\boldsymbol{x}_i)\right)^2 \\ &\leq \frac{m_2}{n} \sum_{i,j} \left(\boldsymbol{\alpha}^{\mathrm{T}} \boldsymbol{f}(\boldsymbol{x}_i)\right)^2 \left(\max_\chi |a_n(\boldsymbol{x})|\right) \\ &= m_2 \cdot \boldsymbol{\alpha}^{\mathrm{T}} \boldsymbol{M}_n \boldsymbol{\alpha} \cdot \max_\chi \|\boldsymbol{f}(\boldsymbol{x})\| \cdot \left\|\boldsymbol{M}_n^{-1} \boldsymbol{b}_n\right\|, \end{aligned}$$

which is $O\left(n^{-1/2}\right)$. Thus all eigenvalues of $\boldsymbol{T}_n$ are $O\left(n^{-1/2}\right)$, hence so is $\boldsymbol{T}_n$ itself and (A.13) becomes

$$\boldsymbol{P}_n = E\left[\psi'_\sigma(\varepsilon)\right] \boldsymbol{M}_n - O_p\left(n^{-1/2}\right),$$

where we invoke the WLLN. Now (A.12) becomes

$$\sqrt{n}\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right) = \left[E\left[\psi'_\sigma(\varepsilon)\right] \boldsymbol{M}_n + O\left(n^{-1/2}\right)\right]^{-1} \cdot \sqrt{n} \tilde{\boldsymbol{z}}_n, \tag{A.14}$$

for

$$\tilde{\boldsymbol{z}}_n = \frac{1}{n} \sum_{i,j} \psi_\sigma\left(Y_j(\boldsymbol{x}_i) - \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{x}_i) \tilde{\boldsymbol{\theta}}_n\right) \boldsymbol{f}(\boldsymbol{x}_i) = \frac{1}{n} \sum_{i,j} \psi_\sigma\left(\varepsilon_{ij} + h_n(\boldsymbol{x}_i)\right) \boldsymbol{f}(\boldsymbol{x}_i).$$

Recalling that $h_n(\boldsymbol{x})$ is $O\left(n^{-1/2}\right)$, and using (A.4), we have

$$\tilde{\boldsymbol{z}}_n = \frac{1}{n}\sum_{i,j}\psi_\sigma\left(\varepsilon_{ij}+h_n(\boldsymbol{x}_i)\right)\boldsymbol{f}(\boldsymbol{x}_i) = \bar{\boldsymbol{z}}_n + \frac{1}{n}\sum_{i,j}\psi'_\sigma(\varepsilon_{ij})\,h_n(\boldsymbol{x}_i)\,\boldsymbol{f}(\boldsymbol{x}_i) + O_p\left(n^{-1}\right),$$

with (and again invoking the WLLN)

$$\begin{aligned}\sqrt{n}\tilde{\boldsymbol{z}}_n - \sqrt{n}\bar{\boldsymbol{z}}_n &= \frac{1}{n}\sum_{i,j}\psi'_\sigma(\varepsilon_{ij})\ \sqrt{n}h_n(\boldsymbol{x}_i)\,\boldsymbol{f}(\boldsymbol{x}_i) + O_p\left(n^{-1/2}\right)\\ &= E\left[\psi'_\sigma(\varepsilon)\right]\int_\chi \sqrt{n}h_n(\boldsymbol{x})\,\boldsymbol{f}(\boldsymbol{x})\,\xi_n(d\boldsymbol{x}) + o_p(1)\\ &= E\left[\psi'_\sigma(\varepsilon)\right]\left\{\sqrt{n}\boldsymbol{b}_n - \sqrt{n}\boldsymbol{M}_n\left(\boldsymbol{M}_n^{-1}\boldsymbol{b}_n\right)\right\} + o_p(1),\end{aligned}$$

which is $o_p(1)$. Thus $\sqrt{n}\tilde{\boldsymbol{z}}_n = \sqrt{n}\bar{\boldsymbol{z}}_n + o_p(1)$ and (A.14) becomes

$$\sqrt{n}\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right) = \left[E\left[\psi'_\sigma(\varepsilon)\right]\boldsymbol{M}_n + O\left(n^{-1/2}\right)\right]^{-1}\left(\sqrt{n}\bar{\boldsymbol{z}}_n + o_p(1)\right),$$

whence

$$\sqrt{n}\boldsymbol{M}_n^{1/2}\left(\hat{\boldsymbol{\theta}}_n - \tilde{\boldsymbol{\theta}}_n\right) = \frac{1}{E\left[\psi'_\sigma(\varepsilon)\right]}\boldsymbol{M}_n^{-1/2}\sqrt{n}\bar{\boldsymbol{z}}_n + o_p(1). \tag{A.15}$$

From this, the statement of the Theorem is immediate. □

A.2. Proof of Theorem 2

To maximize AIMSE over $\tau \in \Upsilon$, note that both of its first two terms at (10) become larger if $\tau(\boldsymbol{x})$ is multiplied by a constant exceeding one in absolute value, hence at a maximum (4) is attained with equality. Define $\tau_0(\boldsymbol{x}) = \sqrt{n}\tau(\boldsymbol{x})/\kappa$ and $\boldsymbol{c}(\xi) = \sqrt{n}\boldsymbol{b}_0(\xi)/\kappa$. Then with $\boldsymbol{c}_n = \boldsymbol{c}(\xi_n)$ we have that $\max_{\tau\in\Upsilon}$ AIMSE$(\xi_n|\tau)$ is

$$J(\xi_n) \stackrel{def}{=} \sigma_M^2 tr\boldsymbol{A}\boldsymbol{M}_n^{-1} + \kappa^2\left(1 + \max_{\tau_0}\boldsymbol{c}_n^{\mathrm{T}}\boldsymbol{M}_n^{-1}\mathbf{A}\boldsymbol{M}_n^{-1}\boldsymbol{c}_n\right), \tag{A.16}$$

with τ_0 – which appears in $\boldsymbol{c}_n$ via $\boldsymbol{b}_0$ – constrained by (3) and by $\sum_{i=1}^N \tau_0^2(\boldsymbol{x}_i) = 1$.

To carry out the maximization in (A.16) recall that a feature of the Gram-Schmidt process used in the construction of $\boldsymbol{Q}$ is that $\boldsymbol{F} = \boldsymbol{Q}\boldsymbol{T}$ for a non-singular lower triangular matrix $\boldsymbol{T}$. Let $\boldsymbol{Q}_\perp : N \times N - p$ be the orthogonal complement of $\boldsymbol{Q}$, so that $\left(\boldsymbol{Q}\vdots\boldsymbol{Q}_\perp\right) : N \times N$ is orthogonal. The condition in (3) asserts that $\boldsymbol{\tau}_0$ is orthogonal to the columns of $\boldsymbol{Q}$, hence is a linear combination of the columns of $\boldsymbol{Q}_\perp$ and so there is $\boldsymbol{\beta}_{N-p\times 1}$ of unit norm for which $\boldsymbol{\tau}_0 = \boldsymbol{Q}_\perp\boldsymbol{\beta}$.

In this notation $\boldsymbol{A} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = \boldsymbol{T}^{\mathrm{T}}\boldsymbol{T}$ and

$$\begin{aligned}\boldsymbol{M}_n &= \boldsymbol{f}^{\mathrm{T}}\boldsymbol{D}(\xi_n)\boldsymbol{F} = \boldsymbol{T}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}(\xi_n)\boldsymbol{Q}\boldsymbol{T},\\ \boldsymbol{c}_n &= \boldsymbol{f}^{\mathrm{T}}\boldsymbol{D}(\xi_n)\boldsymbol{\tau}_0 = \boldsymbol{T}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}(\xi_n)\boldsymbol{Q}_\perp\boldsymbol{\beta};\end{aligned}$$

then (A.16) becomes

$$J\left(\xi_n\right)=\sigma_M^2 tr\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-1}+\kappa^2\left(1+\max_{\|\boldsymbol{\beta}\|=1}\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{Q}_{\perp}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-2}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}_{\perp}\boldsymbol{\beta}\right). \tag{A.17}$$

Noting that the maximum eigenvalue of a matrix $\boldsymbol{P}\boldsymbol{P}^{\mathrm{T}}$ is that of $\boldsymbol{P}^{\mathrm{T}}\boldsymbol{P}$, and that $\boldsymbol{Q}_{\perp}\boldsymbol{Q}_{\perp}^{\mathrm{T}}=\left(\boldsymbol{I}_N-\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}}\right)$, we have

$$\begin{aligned}
&\max_{\|\boldsymbol{\beta}\|=1}\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{Q}_{\perp}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-2}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}_{\perp}\boldsymbol{\beta}\\
=\;& ch_{\max}\boldsymbol{Q}_{\perp}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-1}\cdot\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-1}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}_{\perp}\\
=\;& ch_{\max}\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-1}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}_{\perp}\boldsymbol{Q}_{\perp}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-1}\\
=\;& ch_{\max}\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-1}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}^2\left(\xi_n\right)\boldsymbol{Q}\left(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\xi_n\right)\boldsymbol{Q}\right)^{-1}-1.
\end{aligned}$$

This in (A.17) gives (11). □

A.3. Proof of Lemma 2

Using (8), an asymptotic approximation is

$$\textsc{aim}=\sum_{i=1}^{N}E\left[\rho\left(\tau\left(\boldsymbol{x}_i\right)-\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\left(\boldsymbol{M}_n^{-1}\boldsymbol{b}_n+\left(\sigma_M/\sqrt{n}\right)\boldsymbol{M}_n^{-1/2}\boldsymbol{Z}\right)\right)\right],$$

for a random vector $\boldsymbol{Z}\sim\boldsymbol{N}\left(\boldsymbol{0}_p,\boldsymbol{I}_p\right)$. Define random variables W_i by

$$W_i=\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\left(\sigma_M/\sqrt{n}\right)\boldsymbol{M}_n^{-1/2}\boldsymbol{Z}\sim N\left(0,\left(\sigma_M^2/n\right)\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\boldsymbol{M}_n^{-1}\boldsymbol{f}\left(\boldsymbol{x}_i\right)\right),$$

and set $\alpha_i\left(\boldsymbol{\tau}\right)=\tau\left(\boldsymbol{x}_i\right)-\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\boldsymbol{M}_n^{-1}\boldsymbol{b}_n$. Then

$$\textsc{aim}=\sum_{i=1}^{N}E\left[\rho\left(\alpha_i\left(\boldsymbol{\tau}\right)-W_i\right)\right]. \tag{A.18}$$

To write AIM as a function of $\boldsymbol{\beta}$ we note that, in terms of the orthogonal basis,

$$\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\boldsymbol{M}_n^{-1}\boldsymbol{b}_n=\left(\kappa/\sqrt{n}\right)\left(\boldsymbol{Q}\boldsymbol{R}^{-1}\left(\boldsymbol{\xi}_n\right)\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{D}\left(\boldsymbol{\xi}_n\right)\boldsymbol{Q}_{\perp}\beta\right)_i,$$

whence $\alpha_i\left(\boldsymbol{\tau}\right)=\left(\kappa/\sqrt{n}\right)\boldsymbol{h}_i^{\mathrm{T}}\left(\boldsymbol{\xi}_n\right)\boldsymbol{\beta}$. Since

$$\textsc{var}\left[W_i\right]=\left(\sigma_M^2/n\right)\boldsymbol{f}^{\mathrm{T}}\left(\boldsymbol{x}_i\right)\boldsymbol{M}_n^{-1}\boldsymbol{f}\left(\boldsymbol{x}_i\right)=\left(\sigma_M^2/n\right)\left[\boldsymbol{Q}\boldsymbol{R}^{-1}\left(\boldsymbol{\xi}_n\right)\boldsymbol{Q}^{\mathrm{T}}\right]_{ii},$$

we can write $W_i=\left(\sigma_M/\sqrt{n}\right)s_i\left(\boldsymbol{\xi}_n\right)Z_i$. This in (A.18) gives (15). □

A.4. Proof of Lemma 3

That AIM is given by (16) is immediate from (15). Then the jth element of $\boldsymbol{G}_1(\boldsymbol{\beta})$ is

$$\begin{aligned}
\frac{\partial G(\boldsymbol{\beta})}{\partial \beta_j} &= \sum_{i=1}^{N}\int_{-\infty}^{\infty}\psi\left(\frac{\kappa}{\sqrt{n}}\boldsymbol{h}_i^{\mathrm{T}}(\boldsymbol{\xi}_n)\boldsymbol{\beta}-\frac{\sigma_M}{\sqrt{n}}s_i(\boldsymbol{\xi}_n)z\right)\phi(z)\,dz\frac{\partial \boldsymbol{h}_i^{\mathrm{T}}(\boldsymbol{\xi}_n)\boldsymbol{\beta}}{\partial\beta_j}\\
&= \frac{\kappa}{\sqrt{n}}\sum_{i=1}^{N}\int_{-\infty}^{\infty}\left(\psi\left(\boldsymbol{\eta}(\boldsymbol{\xi}_n,z)\right)\right)_i\phi(z)\,dzH_{ij}\\
&= \frac{\kappa}{\sqrt{n}}\left(\boldsymbol{H}^{\mathrm{T}}(\boldsymbol{\xi}_n)\int_{-\infty}^{\infty}\psi\left(\boldsymbol{\eta}(\boldsymbol{\xi}_n,z)\right)\phi(z)\,dz\right)_j,
\end{aligned}$$

as claimed. The $(j,k)^{th}$ element of $\boldsymbol{G}_2(\boldsymbol{\beta})$ is

$$\begin{aligned}
\frac{\partial\left(\boldsymbol{G}_1(\boldsymbol{\beta})\right)_j}{\partial\beta_k} &= \frac{\partial}{\partial\beta_k}\frac{\kappa}{\sqrt{n}}\sum_{i=1}^{N}\left(\boldsymbol{H}^{\mathrm{T}}(\boldsymbol{\xi}_n)_{ji}\right)\int_{-\infty}^{\infty}\left(\psi\left(\boldsymbol{\eta}(\boldsymbol{\xi}_n,z)\right)\right)_i\phi(z)\,dz\\
&= \frac{\kappa^2}{n}\sum_{i=1}^{N}\left(\boldsymbol{H}^{\mathrm{T}}(\boldsymbol{\xi}_n)_{ji}\right)\int_{-\infty}^{\infty}\left(\psi'\left(\boldsymbol{\eta}(\boldsymbol{\xi}_n,z)\right)\right)_i\phi(z)\,dz\frac{\partial\boldsymbol{h}_i^{\mathrm{T}}(\boldsymbol{\xi}_n)\boldsymbol{\beta}}{\partial\beta_k}\\
&= \frac{\kappa^2}{n}\sum_{i=1}^{N}\left(\boldsymbol{H}^{\mathrm{T}}(\boldsymbol{\xi}_n)_{ji}\right)\int_{-\infty}^{\infty}\left(\psi'\left(\boldsymbol{\eta}(\boldsymbol{\xi}_n,z)\right)\right)_i\phi(z)\,dzH_{i,k}\\
&= \frac{\kappa^2}{n}\boldsymbol{H}^{\mathrm{T}}(\boldsymbol{\xi}_n)\oplus_{i=1}^{N}\left[\int_{-\infty}^{\infty}\psi'\left(\boldsymbol{\eta}(\boldsymbol{\xi}_n,z)\right)\phi(z)\,dz\right]_i\boldsymbol{H}(\boldsymbol{\xi}_n)
\end{aligned}$$

as claimed. □

A.5. Proof of Lemma 4

We calculate that, using the Huber score function,

$$\frac{\sigma_M^2}{\sigma^2}=\frac{1-2c\phi(c)+2\left(c^2-1\right)\Phi(-c)}{\left(1-2\Phi(-c)\right)^2}\overset{def}{=}G(c),$$

in terms of which $\nu_{\mathrm{M}}=\left(\gamma^2G(c)+1\right)^{-1}$. The function G is the asymptotic variance of the M-estimate of location with $N(0,1)$ errors. It is decreasing in c and must exceed the asymptotic variance of the sample mean: $G(c)>G(\infty)=1$. Two applications of L'Hôpital's rule give $G(0)=\pi/2$, the asymptotic variance of the median of a normal sample, with corresponding regression estimate given by $\psi(x)=sign(x)$. Thus

$$0<\nu_{\mathrm{LS}}-\nu_{\mathrm{M}}=\frac{\gamma^2}{\gamma^2+1}\left\{\frac{G(c)-1}{\gamma^2G(c)+1}\right\}\overset{def}{=}H\left(c,\gamma^2\right).$$

For fixed γ, $H(c,\gamma^2)$ is increasing in $G(c)$, so is maximized at $c=0$. Then $H(0,\gamma^2)$ vanishes at $\gamma^2=0,\infty$ and is maximized at $\gamma^2=[G(0)]^{-1/2}$, with value $\max_{c,\gamma}(\nu_{\mathrm{LS}}-\nu_{\mathrm{M}})=\left(\sqrt{G(0)}-1\right)\Big/\left(\sqrt{G(0)}+1\right)$. At the maximum, $\nu_{\mathrm{LS}}=1/\left(\gamma^2+1\right)$ and $\nu_{\mathrm{M}}=1/\left(\gamma^2G(0)+1\right)$, giving (19) . □

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